

Considering an [Optimization Problem] with assumptions $\mathcal{D} \neq \emptyset$, and optimal value p^* exists. The *Lagrange dual function* (or *dual function*) $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$, where $\lambda \in \mathbb{R}^m$ and $\nu \in \mathbb{R}^p$ as

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x) \right).$$

where $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ is the [Lagrangian] associated with the [Optimization Problem]. To note, since the dual function is the pointwise infimum of a family of affine functions of (λ, ν) , it is concave.

0.0.1 See also

0.0.2 References